

CURRENCY CONVERTER

API Documentation

FUTM-SWE-221 · Project 6 · Developed by Aisha

API: Open Exchange Rates · openexchangerates.org

App ID: 9069645b9f1f4dc4bb634627a5e6d9d4

Project Overview

This is Project 6 of the FUTM-SWE-221 Individual Software Development Project assignment. The application is a Currency Converter that integrates the Open Exchange Rates REST API to fetch live exchange rates for five currencies: NGN, USD, EUR, GBP, and GHS. The free plan base is always USD. Conversion formula: $\text{converted} = \text{amount} \times (\text{rate_TO} / \text{rate_FROM})$.

SDLC Phases

Phase	Details
Planning	Identified five currencies for Nigeria. Selected Open Exchange Rates as API provider (free plan, 170+ currencies).
Analysis	End user is a market trader who needs quick NGN conversions on mobile. No registration — just open and use.
Design	Single-page mobile-first UI with a large result display and a swap button for reversing currency pairs quickly.
Implementation	PHP and cURL call <code>/latest.json</code> . Formula: $\text{converted} = \text{amount} \times (\text{rate_TO} / \text{rate_FROM})$. Base is always USD.
Testing	Tested NGN→USD, USD→NGN, NGN→GHS, EUR→GBP, zero amounts, same-currency pairs, and API timeouts.
Deployment	Single PHP file on any server with cURL. No database or framework needed. Runs immediately on any mobile device.

API Information

Property	Value
Provider	Open Exchange Rates
Base URL	https://openexchangerates.org/api
App ID	9069645b9f1f4dc4bb634627a5e6d9d4
Authentication	Query param: <code>app_id=YOUR_APP_ID</code>
Response Format	JSON
Free Plan Quota	1,000 requests per month. Hourly rate updates.
Base Currency	Always USD on free plan. Custom base requires a paid plan.
Conversion Formula	$\text{converted} = \text{amount} \times (\text{rate_TO} / \text{rate_FROM})$

Property	Value
Documentation	https://docs.openexchangerates.org

Test Currencies

Code	Currency	Symbol	Rate / 1 USD	Role
NGN	Nigerian Naira	₦	1,370.500000	Base / Anchor
USD	US Dollar	\$	1.000000	Target
EUR	Euro	€	0.880038	Target
GBP	British Pound	£	0.757958	Target
GHS	Ghanaian Cedi	GH¢	11.225000	Target

API Requests and Responses

All five requests below were sent to the Open Exchange Rates API. The request URL, parameters, and full JSON response body are shown for each test.

1. Test 1: Check API Usage and Quota

This request checks how many of the 1,000 monthly free requests have been used and confirms the App ID is active.

Request Details:

Method : GET
URL : https://openexchangerates.org/api/usage.json?app_id=9069645b9f1f4dc4bb634627a5e6d9d4
Param : app_id = 9069645b9f1f4dc4bb634627a5e6d9d4 (Required: Yes)
Status : 200 OK

Response Body:

```
{
  "status": 200,
  "data": {
    "app_id": "9069645b9f1f4dc4bb634627a5e6d9d4",
    "status": "active",
    "plan": {
      "name": "Free",
      "quota": "1000 requests / month",
      "update_frequency": "3600s"
    },
    "usage": {
      "requests": 0,
      "requests_quota": 1000,
      "requests_remaining": 1000,
      "days_elapsed": 0,
      "days_remaining": 30,
      "daily_average": 0
    }
  }
}
```

2. Test 2: Get Currency Names List

Retrieves human-readable names for all 170+ supported currencies. Public endpoint — no App ID required.

Request Details:

Method : GET
URL : <https://openexchangerates.org/api/currencies.json>
Param : none (Public endpoint – no authentication required)
Status : 200 OK

Response Body:

```
{
  "AED": "UAE Dirham",
  "EUR": "Euro",
  "GBP": "Pound Sterling",
  "GHS": "Ghanaian Cedi",
  "NGN": "Nigerian Naira",
  "USD": "United States Dollar"
}
```

3. Test 3: Get Latest Rates for All 5 Currencies

Fetches live exchange rates for all five test currencies. The base is always USD on the free plan. Rates are updated every hour.

Request Details:

Method : GET
URL : https://openexchangerates.org/api/latest.json?app_id=9069645b9f1f4dc4bb634627a5e6d9d4&symbols=NGN,USD,EUR,GBP,GHS
Param : app_id = 9069645b9f1f4dc4bb634627a5e6d9d4 (Required: Yes)
Param : symbols = NGN,USD,EUR,GBP,GHS (Required: No)
Status : 200 OK

Response Body:

```
{
  "disclaimer": "Usage subject to terms: https://openexchangerates.org/terms",
  "license": "https://openexchangerates.org/license",
  "timestamp": 1782277200,
  "base": "USD",
  "rates": {
    "EUR": 0.880038,
    "GBP": 0.757958,
    "GHS": 11.225,
    "NGN": 1370.5,
    "USD": 1
  }
}
```

4. Test 4: Convert 10,000 NGN to USD

Converts 10,000 Nigerian Naira (₦) to US Dollars. Formula: converted = amount × (rate_TO / rate_FROM). Result: 10000 × (1 / 1370.5) = 7.2969 USD

Request Details:

Method : GET
URL : https://openexchangerates.org/api/latest.json?app_id=9069645b9f1f4dc4bb634627a5e6d9d4&symbols=NGN,USD
Param : app_id = 9069645b9f1f4dc4bb634627a5e6d9d4 (Required: Yes)
Param : symbols = NGN,USD (Required: No)
Status : 200 OK

Response Body:

```
{
  "disclaimer": "Usage subject to terms: https://openexchangerates.org/terms",
  "license": "https://openexchangerates.org/license",
  "timestamp": 1782277200,
  "base": "USD",
  "rates": {
    "NGN": 1370.5,
    "USD": 1
  }
}
```

Calculation:

```
amount      = 10,000 NGN (₦)
rate_FROM   = 1370.5
rate_TO     = 1
result      = 10000 × (1 / 1370.5) = 7.2969 USD
```

5. Test 5: Convert 1 USD to NGN (Reverse Pair)

Converts 1 US Dollar back to Nigerian Naira (₦). Formula: converted = amount × (rate_TO / rate_FROM).

Result: $1 \times (1370.5 / 1) = 1,370.50$ NGN

Request Details:

Method : GET
URL : https://openexchangerates.org/api/latest.json?app_id=9069645b9f1f4dc4bb634627a5e6d9d4&symbols=NGN,USD
Param : app_id = 9069645b9f1f4dc4bb634627a5e6d9d4 (Required: Yes)
Param : symbols = NGN,USD (Required: No)
Status : 200 OK

Response Body:

```
{
  "disclaimer": "Usage subject to terms: https://openexchangerates.org/terms",
  "license": "https://openexchangerates.org/license",
  "timestamp": 1782277200,
  "base": "USD",
  "rates": {
    "NGN": 1370.5,
    "USD": 1
  }
}
```

Calculation:

```
amount      = 1 USD
rate_FROM   = 1
rate_TO     = 1370.5
result      = 1 × (1370.5 / 1) = 1,370.50 NGN (₦)
```

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